Cheatsheet AZ-104 — 04 · Storage accounts: types, redundancy, encryption, object replication, firewall, Storage Explorer and AzCopy

1. Account types

Type	Services	Redundancy	When
Standard general-purpose v2 (StorageV2)	Blob (incl. Data Lake Gen2), Files, Queue, Table	LRS, ZRS, GRS, RA-GRS, GZRS, RA-GZRS	The default option for almost everything; the only one with **geo-redundancy** and **access tiers**
Premium block blobs (BlockBlobStorage)	Block and append blobs	LRS, ZRS	Many transactions, small objects, consistently low latency. **No tiers**
Premium file shares (FileStorage)	Azure Files only: SSD provisioned v1 / v2 (and HDD provisioned v2)	LRS, ZRS (HDD v2 also GRS / GZRS)	Files with high IOPS; the only type with **NFS 4.1**
Premium page blobs	Page blobs only	LRS (ZRS in some regions)	Unmanaged disks / VHD; rare today

- **Legacy:** General-purpose v1 and Blob Storage are being retired → **upgrade to GPv2** (no downtime, **irreversible**). Classic (ASM) accounts were retired in August 2024. The type cannot be changed after creation: create another account and copy.
- **Name:** 3-24 characters, lowercase letters and numbers only, **unique across all of Azure**. Endpoints: `https://<account>.blob.core.windows.net`, `.file.`, `.queue.`, `.table.`, `.dfs.` (Data Lake) and `.z<n>.web.` (static website, `$web` container).
- Default limits: **250** accounts per region and subscription (500 on request), **5 PiB** per account, 20,000-40,000 requests/s, 400 IP rules and 400 VNet rules per account.
- At creation: performance (Standard / Premium), redundancy, default **access tier** (Hot / Cool / Cold), **hierarchical namespace** (Data Lake: real folders, POSIX ACLs; can be enabled later but not disabled), secure transfer (HTTPS) on, minimum TLS 1.2, anonymous blob access **disabled**, key access allowed, **infrastructure encryption** (only at creation), **large file shares**.

2. Redundancy

Option	Copies	Durability	Survives	Read from secondary
LRS	3 in one datacenter	11 nines	Disk, node or rack failure	—
ZRS	3 across **3 zones** in the region	12 nines	Datacenter / zone outage	—
GRS	LRS + 3 **asynchronous** LRS copies in the paired region	16 nines	Regional outage (with failover)	No
RA-GRS	GRS + `-secondary` endpoint	16 nines	Same, and also **reads** without failover	**Yes**
GZRS	ZRS + 3 LRS copies in the paired region	16 nines	Zone and region	No
RA-GZRS	GZRS + read	16 nines	Zone and region, read always	**Yes**

- **Restrictions:** Premium only LRS / ZRS. **Azure Files does not support RA-GRS or RA-GZRS** (it never reads from the secondary). The **Archive** tier does not exist in ZRS / GZRS / RA-GZRS. Geo copies are **asynchronous**: RPO ≈ 15 min (**Last Sync Time** shows how far replication has reached).
- **Changing redundancy:** most changes are made **on the account itself** (Settings → Redundancy): LRS ↔ GRS / RA-GRS are immediate; LRS ↔ ZRS, GRS ↔ GZRS are a **conversion** (customer-initiated, can take days; or open a ticket). If there are blobs in Archive they must be rehydrated before moving to ZRS / GZRS. There is no Standard ↔ Premium conversion: new account + copy (AzCopy).
- **Failover:** with GRS / GZRS you can force a **customer-managed failover** (unplanned) when the primary region goes down: the secondary becomes primary, the account ends up as **LRS** and **GRS must be reconfigured**; unreplicated data may be lost. There is also the **planned failover** (no loss, for drills). With RA-GRS it is enough to read from the `-secondary` endpoint without failover. Failover is not possible with object replication enabled, nor with Azure File Sync without support.

3. Encryption

Mechanism	What it is	Detail
SSE with Microsoft-managed keys	AES-256 encryption at rest **always on**, no cost or configuration	Cannot be disabled
Customer-managed keys (CMK)	Your key in **Key Vault** (or Managed HSM) + the account's **managed identity**	The vault needs **soft delete and purge protection**; automatic rotation if you reference the key without a version; revoking the key = data inaccessible
Customer-provided keys	The client sends the key in each request (blobs only)	For apps that manage their own keys; not for the general exam
Infrastructure (double) encryption	Second encryption layer at the infrastructure level	Only enabled **at account creation**
Encryption scopes	Different keys per container or blob within the same account	Isolate data from several customers in one account
In transit	**Secure transfer required** (HTTPS and encrypted SMB 3.x only), minimum TLS	Disable it only for legacy SMB 2.1 (and only within the region)

- Access keys, SAS and RBAC are **authorization**, not encryption. Azure Disk Encryption and encryption at host belong to **VM disks**, not to the account (cheatsheet 08).

4. Object replication (blobs)

Aspect	Rule
What it copies	**Block blobs** (not append, not page, not snapshots) from a source container to a destination one, **asynchronously**, with their versions and metadata
Requirements	**Blob versioning** on source and destination, **change feed** on source; GPv2 or Premium block blob accounts; **no** hierarchical namespace
Topology	Same or different region, same or different subscription, even another tenant (if allowed). One source → maximum **2** destination accounts; **1,000 rules** per policy
Filters	Name prefix and minimum creation date; by default only **new** blobs (existing ones can be included)
Destination	Becomes **read-only** while a policy exists (writing returns 409); an already replicated blob can be archived at the destination, but its later changes no longer replicate
Purpose	Reduce latency per region, process in several regions, distribute results, **compliance** / data copy; **does not replace GRS** (it is per container and you control it)
Breaks if	A source or destination blob moves to **Archive**, there is an immutability policy on the destination, or account failover is used

5. Account firewall and networking

Setting	Options	Detail
Public network access	Enabled from all networks · Enabled from **selected virtual networks and IP addresses** · **Disabled**	Disabled = private endpoints only
Virtual network rules	Subnets with the `Microsoft.Storage` **service endpoint** (same region) or `Microsoft.Storage.Global` (any region)	Traffic arrives with a **private** source IP; up to 400 rules; VNets from any subscription or tenant
IP network rules	**Public** IPv4 addresses: a single IP or a CIDR range of **/30 or larger** (not /31 or /32, no private ranges); they do not work for clients in the same Azure region (those use VNet rules)	For on-prem: the public egress IP (or the ExpressRoute Microsoft peering NAT IPs); up to 400
Resource instance rules	Allow specific Azure resources (Synapse, Backup vaults, etc.) by identity	For services that cannot be isolated by IP / VNet
Exceptions	"Allow Azure services on the **trusted services list**" · read access to **resource logs** · read access to **metrics**	Backup, Monitor, Defender, Site Recovery... keep working with the network closed
Private endpoint	NIC with a private IP in your VNet per sub-resource (blob, file, dfs, queue, table, web) + private zone `privatelink.<service>.core.windows.net`	Access from on-prem via VPN / ExpressRoute and from peered VNets; it is the answer to "no public endpoint" (cheatsheet 13)

- With the firewall enabled, the **portal, Storage Explorer and AzCopy from your PC** fail ("not authorized") until you add your **public IP** to the rules (or use "Add your client IP address").
- **Service endpoint** = still the public endpoint, but it only accepts those subnets; free; does not work from on-prem. **Private endpoint** = private IP, private DNS, cost per hour and per GB; works from on-prem.
- Other account switches: **Allow Blob anonymous access** (if Disabled no container can be public), **Allow storage account key access** (Disabled = Entra ID and user delegation SAS only), **Default to Microsoft Entra authorization in the portal**, **Allow cross-tenant replication**.

6. Storage Explorer, AzCopy and bulk transfer

Tool	How it authenticates	Purpose
Azure Storage Explorer (desktop app)	Subscription (Entra ID), name + key, **SAS URI**, connection string, emulator	Browse and manage blobs, files, queues and tables; upload / download; change tier; create SAS and stored access policies; Data Lake ACLs; view soft-deleted items; share snapshots. To view data with Entra ID it needs the **Storage Blob Data Reader** role (plain Reader does not see data)
AzCopy (command line)	`azcopy login` (Entra ID), `--identity` (managed identity) or **SAS in the URL**; Files: SAS (or Entra with the Storage File Data Privileged Contributor role)	Bulk, scriptable copies: local ↔ blob / Files, **blob ↔ blob** between accounts (server to server), **S3 / Google Cloud → Azure**, incremental sync, resume jobs
Azure Import/Export	Your own BitLocker-encrypted SATA/SSD drives + the WAImportExport tool	Import to Blob or Files, export Blob only; when you do not want Microsoft hardware
Data Box Disk / Data Box 120 / Data Box 525	Microsoft devices (up to 40 TB on SSD drives · **120 TB** usable · **525 TB** usable; the 80 TB Data Box and Data Box Heavy are retired or being retired)	Tens or hundreds of TB with little bandwidth; Data Box Gateway = **online** virtual appliance
Azure Storage Mover	Managed service	Migrate on-prem SMB / NFS shares to Azure Files or Blob over the network

```
azcopy login                                  # Entra ID (or --identity on a VM with MI)
azcopy copy "C:\data" "https://st1.blob.core.windows.net/cont?<SAS>" --recursive
azcopy copy "https://src.blob.core.windows.net/c1?<SAS>" "https://dst.blob.core.windows.net/c2?<SAS>" --recursive
azcopy sync "C:\data" "https://st1.blob.core.windows.net/cont?<SAS>" --recursive --delete-destination=true
azcopy copy ... --block-blob-tier Cool --include-pattern "*.pdf" --put-md5 --cap-mbps 100
azcopy make "https://st1.blob.core.windows.net/new?<SAS>"     · azcopy list ... · azcopy remove ...
azcopy jobs list · azcopy jobs resume <jobId>                # resume an interrupted copy
```

- `copy` overwrites by default; `sync` is **one-way** and incremental (compares last-modified time) and only deletes at the destination with `--delete-destination`. **Account-to-account** copies do not go through your PC. For Azure Files, AzCopy requires SAS except in recent versions with Entra ID.

- Choosing: a few GB or TB with good bandwidth → AzCopy / Storage Explorer; tens of TB and little network → Data Box Disk / Data Box 120; hundreds of TB → Data Box 525; "no third-party hardware", "use our own drives" → Import/Export; continuous migration of a file server → Azure File Sync.

7. Commands that show up

Task	PowerShell (Az.Storage)	Azure CLI
Create account	`New-AzStorageAccount -ResourceGroupName RG1 -Name st1 -Location westeurope -SkuName Standard_GRS -Kind StorageV2 -AccessTier Hot -MinimumTlsVersion TLS1_2 -AllowBlobPublicAccess $false`	`az storage account create -n st1 -g RG1 -l westeurope --sku Standard_GRS --kind StorageV2 --access-tier Hot --min-tls-version TLS1_2 --allow-blob-public-access false`
Change redundancy or tier	`Set-AzStorageAccount -ResourceGroupName RG1 -Name st1 -SkuName Standard_GRS` · `-AccessTier Cool` · to ZRS / GZRS (conversion): `Start-AzStorageAccountMigration -ResourceGroupName RG1 -AccountName st1 -TargetSku Standard_ZRS`	`az storage account update -n st1 -g RG1 --sku Standard_RAGRS` · `--access-tier Cool` · to ZRS / GZRS: `az storage account migration start -n st1 -g RG1 --sku Standard_ZRS`
Upgrade GPv1 → GPv2	`Set-AzStorageAccount -ResourceGroupName RG1 -Name st1 -UpgradeToStorageV2`	`az storage account update -n st1 -g RG1 --set kind=StorageV2`
Firewall	`Update-AzStorageAccountNetworkRuleSet -ResourceGroupName RG1 -Name st1 -DefaultAction Deny -Bypass AzureServices` · `Add-AzStorageAccountNetworkRule ... -IPAddressOrRange 203.0.113.0/24` · `-VirtualNetworkResourceId <subnet id>`	`az storage account update -n st1 --default-action Deny --bypass AzureServices` · `az storage account network-rule add -n st1 --ip-address 203.0.113.0/24` · `--vnet-name vnet1 --subnet sub1`
Failover	`Invoke-AzStorageAccountFailover -ResourceGroupName RG1 -Name st1`	`az storage account failover -n st1 -g RG1`
Endpoints and keys	`(Get-AzStorageAccount -ResourceGroupName RG1 -Name st1).PrimaryEndpoints` · `Get-AzStorageAccountKey`	`az storage account show -n st1 --query primaryEndpoints` · `az storage account keys list -n st1`
Object replication	`New-AzStorageObjectReplicationPolicy -ResourceGroupName RG1 -StorageAccountName dst -SourceAccount src -Rule <rule>`	`az storage account or-policy create --account-name dst --source-account src --source-container c1 --destination-container c2`
PowerShell context	`$ctx = New-AzStorageContext -StorageAccountName st1 -UseConnectedAccount` (Entra) · `-StorageAccountKey <key>` · `-SasToken <sas>`	CLI: `--auth-mode login` (Entra) · `--account-key` · `--sas-token` · `--connection-string`

8. Decision tree

```
What does the wording ask for?
├─ Blobs, queues, tables and files, geo-redundancy, tiers ..... Standard general-purpose v2
├─ NFS, high IOPS on files ..... Premium file shares (FileStorage)
├─ Many small transactions, minimal latency on blobs ..... Premium block blobs
├─ Survive a datacenter outage within the region ..... ZRS
├─ Survive a regional outage and be able to READ meanwhile ..... RA-GRS / RA-GZRS
├─ Survive region + zone ..... GZRS
├─ Data that cannot leave the region ..... LRS or ZRS
├─ Copy specific containers to another account or region ..... Object replication (versioning + change feed)
├─ Encryption with our own key ..... CMK in Key Vault + managed identity
├─ Double encryption required by compliance ..... Infrastructure encryption (at creation)
├─ Only reachable from our VNet ..... Firewall: selected networks + service endpoint
├─ No public endpoint, access from on-prem ..... Private endpoint + private DNS
├─ Backup / Monitor must keep working with the firewall closed ..... Exception: trusted Microsoft services
├─ Copy TB over the network in a scriptable way ..... AzCopy
└─ Hundreds of TB with little bandwidth ..... Data Box 120 / 525, or Import-Export with your own drives
```

9. Signals → answer

Wording	Answer
"minimize cost", "data can be reconstructed"	LRS
"survive a datacenter outage", "zone failure"	ZRS (or GZRS if the region too)
"read access during a regional outage"	RA-GRS / RA-GZRS
"Azure Files", "read from secondary region"	Impossible: Files does not support RA-*; use GRS + failover
"archive tier" + "ZRS"	Incompatible: Archive only LRS / GRS / RA-GRS
"change from LRS to GRS"	Settings → Redundancy (no recreation)
"change from Standard to Premium"	New account + AzCopy
"after failover the account is LRS"	Normal: reconfigure GRS afterwards
"NFS file share"	Premium file shares (FileStorage)
"hierarchical namespace", "ACLs on folders", "analytics"	Data Lake Storage Gen2 (GPv2 + HNS)
"replicate blobs from container A to container B in another region"	Object replication
"object replication fails", "destination 409"	Missing versioning / change feed, blob in Archive, or writing to the destination
"encrypt with keys we control", "rotate keys"	Customer-managed keys in Key Vault
"storage not accessible from the portal after enabling firewall"	Add the client's public IP (or trusted services for services)
"VM in a VNet must reach storage, no public access from internet"	Service endpoint + VNet rule (or private endpoint)
"on-premises access through private IP"	Private endpoint + VPN / ExpressRoute + DNS

Wording	Answer
"Azure Backup fails after restricting network"	Exception "trusted Microsoft services"
"copy 50 TB, 10 Mbps link"	Data Box
"sync a local folder to a container daily"	`azcopy sync` (scheduled)
"copy between two storage accounts without downloading"	`azcopy copy` account to account (with SAS on both sides)
"Storage Explorer: Reader cannot see blobs"	Add Storage Blob Data Reader or use the key

10. Mistakes to avoid

1. Choosing RA-GRS for Azure Files or Archive on a ZRS account.
2. Believing GRS allows reading from the secondary without failover: that is **RA**-GRS.
3. Thinking failover leaves the account as GRS: it ends up as LRS and must be reconfigured.
4. Trying to convert a Standard account to Premium or change its type: new account + copy.
5. Configuring object replication without versioning on both accounts and change feed on the source, or writing to the destination container.
6. Enabling infrastructure encryption or hierarchical namespace without knowing that one can only be enabled at creation and the other cannot be removed.
7. Closing the firewall and forgetting the trusted services exception (Backup, Monitor) or the client IP.
8. Using `azcopy sync` expecting it to be bidirectional, or `azcopy copy` expecting it to delete what no longer exists at the source.
9. Sending 500 TB over a slow line instead of using Data Box.